

CMPE2600 Lab 1 – Chem Plant – Proving the Turbine Meter

Name: _____

Introduction

Flow measurement equipment frequently needs to have its accuracy verified: this is called ‘proving’. The manufacturer of our turbine meters has provided us with a K-factor for each unit that was determined at the factory for a given set of conditions. (See the brass tag on your Chemical Plant) This exercise requires that you verify the K-factor for your turbine meter.

System Description

The system to be controlled consists of the following digital input and output using the CompactLogix 5370 - 1769-L24ER-QBFC1B referred in document as PLC :

Pump (Digital Output): turns on the AC pump motor when energized.

Drain Valve (Digital Output): Controls drain of main vessel to holding tank. Open when energized.

Drain Valve (Manual): Used when testing to manual drain of main vessel to holding tank.

Start (Digital Input): Used to start the batch process

Stop (Digital Input): Used for emergency stop

Prove (Digital Input): signals a proving run is to begin.

Good (Digital Input): signals a ‘good’ proving run.

Done (Digital Input): signals ‘done’ proving.

Zero (Digital Input): signals the high-speed counter should be zeroed. (High-speed Counter

Quarter_Btn (Digital Input): Valve position control. Quarter of span

Half_Btn (Digital Input): Valve position control. Half of span

Full_Btn (Digital Input): Valve position control. full of span

High Speed Counter (HSC): Signal from the flow meter indicating ticks per gallon. These flow meters have a K factor. This will be wired to one of the 4 HSC available on the PLC. There are two documents to help with HSC, *CompactLogix 5370 Controllers User Manual*, and *1769-um006_-en-*.

The system will also make use of two analog I/O:

Fill Valve (Analog Output 4-20ma): Used to control rate of fill to main Vessel. This will be controlled by the PLC via programming. We will set the Analog output to use raw values giving a that was calculated in previous ICA for the 4-20ma output signal.

Differential Pressure Transducer (Analog Input 4-20ma). The differential pressure transducer is used to indicate the liquid level between the two points on the chem plant. This will use raw values giving a span that was calculated in previous ICA for the 4.20ma input signal

The operating characteristics are as follows:

On ‘start-up’ the system should do the necessary initialization activities and then enter the Waiting state, until the Prove input becomes true to signal the start of a proving run. While in the Waiting state insure that: the pump is off, drain valve is open, and the fill valve is closed.

When you are ready to enter the Proving state, make sure the manual Drain Valve (DV10) is closed, and either manually or automatically close the electrically controlled valve (DV11). The HSC should be reset and enabled once prove is pressed.

Upon Prove becoming momentarily true the system should enter the Proving state. Since the tank is empty, this would be a good time to capture the LevelMin value from the differential pressure sensor. The drain valve takes about 10 seconds to close. Do not start the pump until the drain valve is fully closed. Then the pump will be turned on and the fill valve will be controlled by three buttons assigned to be quarter, half, and full which will represent preset fill valve positions. Prove will start with the fill valve on Full preset.

The 4-20 mA signal will be setup with Raw 16 bit signed integer (-32768-32767) which maps to the 4-20 mA. Setup tags Full, Half, and Quarter accordingly from the span previously calculated in ICA 20.

When a proving run is done and the fill level has been accurately obtained (60 cm), the Good input will signal a move to the Calculate state where the turbine K-factor will be calculated in pulses per US gal. After this the new K-factor should be written to the default K Factor tag. Also, the LevelMax should be captured at this time, before done is pressed.

From the Calculate state, the Done input will signal a move to the Waiting state where the pump is off and the fill valve is closed and the drain valve is open.

At any time in the waiting state, activating the Zero switch will zero the high speed counter.

Emergency Stop

If Stop is pressed at any time the system will shut stop the Pump and Mixer if running, close the fill valve, and ensure the drain is closed, essentially halting everything. The Status should read something like “Emergency Stop Pressed: Press Drain to resume “. Pressing Drain will give the time needed to drain 60cm or thereabouts. Once the drain time is done, the system will return to the waiting state.

Suggestions for Flow

It is suggested to use a Main Ladder program with an entry something like this.



The tag S:FS indicates first scan, which starts a timer. The timer is then held by itself during timing of the timer, allowing that rung to be energized for the length of the timer. The Init rung will be used to set default tag values, ensure everything is off to start, park sequences that are not being used or anything else you might need.

Sequential Function chart is suggested for most of the routines including prove as it suits a state machine approach for actions that are repetitive and needing specific things to happen in a specific

order. The main ladder routine will be used to control the many operations that are needed for the chem plant. You will likely need other tags then the ones mentioned below for operation such as: If you have a Prove, Start, Reverse and Stop as the main options at the waiting state, it would be likely that a tag that indicates what is active ex. ProveEnabled (type bool).

Use a MainMessage String tag to :

- Main menu message that will say press prove to start in Lab1, but will later be modified to include the options available.
- Will have a drain closing message with the timer count down showing the count down
- When filling the MainMessage should be showing the HSC with Press Good when @ 60cm
- Then MainMessage will update with the K factor calculated, Press Done to continue
- MainMessage updates back to Start message.
- MainMessages for Emergency Stop.

State Diagram

Logic Equations

Show the state transition equations and the output equations for your state diagram.

Implementation

Create tags for Prove:

Tag Names	Data Type / IO or Memory / Purpose	Physical IO Address
K-factor	Integer : Calculated at the end of prove	
DefaultK-factor	Integer: Once Prove is done, put in the K-factor Calculated, will be different between machines.	
HSC1_CurrentCount	Integer: Map to high speed counter in program	Local:3:I.Ctr0CurrentCount
Status	String: This will be used to track what is happening currently in the chem plant. ex. “ Press Prove, Start, or Reverse “ would be the waiting screen. Ex. Prove pressed “Proving: Closing drain valve” then “ Proving: Filling . Use button quarter, half and full to control the flow. Press Good @ 60cm .”	
Pump	Bool: Mapped to Physical IO Pump	Local:1:O.Data.8
DrainValve	Bool: Mapped to Physical IO DrainValve	Local:1:O.Data.10
Mixer	Bool: Mapped to Physical IO Mixer	Local:1:O.Data.9
Quarter	Integer: Value to write to VavlePosition	
Half	Integer: Value to write to VavlePosition	
Full	Integer: Value to write to VavlePosition	
ValvePosition	Integer: Mapped to Physical IO Valve	Local:2:O.Ch1.Data
Quarter_Btn	Bool: Mapped to Physical IO Button	Local:1:I.Data.X
Half_Btn	Bool: Mapped to Physical IO Button	Local:1:I.Data.X
Full_Btn	Bool: Mapped to Physical IO Button	Local:1:I.Data.X
Countfor60cm	Integer: Captures HSC during prove when Good is pressed	
Good	Bool: Mapped to Physical IO Button	Local:1:I.Data.X
Done	Bool: Mapped to Physical IO Button	Local:1:I.Data.X
LevelMin	Integer: value stored during prove from differential pressure sensor	
LevelMax	Integer: value stored during prove from differential pressure sensor	
Drain	Bool: Mapped to Physical IO Button	Local:1:I.Data.X

Note: Do NOT mark up the PLC Trainer front panel!

It is suggested that you use Seq2 here, since you will later be using Seq1 along with the code here.

Implement, simulate, and test your solution on the PLC to the extent practical. When you find it necessary connect your PLC to a Chemical Plant for final testing.

When your code is working and **properly documented** ask for sign-off.

Attach your ladder logic printout and the printout of your custom functions to this document.

Sign-off: _____



